

Name: Erik Gonzalez
Date: November 11, 2025
Class: Math 262 – Linear Algebra
Professor: Jerry Rosen

Math 262 – HW/Quiz 4

1. If

$$A = \begin{pmatrix} 4 & 1 \\ -2 & 1 \end{pmatrix}$$

a) Find the characteristic polynomial $\det(A - \lambda I)$ of A .

Work:

$$\begin{aligned} A - \lambda I &= \begin{pmatrix} 4 - \lambda & 1 \\ -2 & 1 - \lambda \end{pmatrix}, \quad \det(A - \lambda I) = (4 - \lambda)(1 - \lambda) - (-2)(1) \\ &= (4 - \lambda)(1 - \lambda) + 2 = 4 - 4\lambda - \lambda + \lambda^2 + 2 = \lambda^2 - 5\lambda + 6. \end{aligned}$$

$$\boxed{\chi_A(\lambda) = \lambda^2 - 5\lambda + 6 = (\lambda - 2)(\lambda - 3)}$$

b) Find the eigenvalues of A by solving the characteristic equation.

$$\lambda^2 - 5\lambda + 6 = 0 \implies (\lambda - 2)(\lambda - 3) = 0.$$

$$\boxed{\lambda_1 = 2, \quad \lambda_2 = 3}$$

c) Give a basis for $\ker(A - \lambda I)$ for each eigenvalue.

For $\lambda = 2$:

$$A - 2I = \begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix} \sim \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix}.$$

$$2x + y = 0 \implies y = -2x, \text{ eigenvector } v_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

$$\ker(A - 2I) = \text{span}\{(1, -2)^\top\}.$$

For $\lambda = 3$:

$$A - 3I = \begin{pmatrix} 1 & 1 \\ -2 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

$$x + y = 0 \implies y = -x, \text{ eigenvector } v_3 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$\ker(A - 3I) = \text{span}\{(1, -1)^\top\}.$$

$$\boxed{\ker(A - 2I) = \text{span}\{(1, -2)^\top\}, \quad \ker(A - 3I) = \text{span}\{(1, -1)^\top\}.$$

d) Provide an eigenbasis of $\mathbb{R}^2$ for A , showing A is diagonalizable.

$v_2 = (1, -2)^\top$ and $v_3 = (1, -1)^\top$ are linearly independent.

$$\boxed{\text{Eigenbasis: } \{(1, -2)^\top, (1, -1)^\top\}}$$

e) Give matrices B and S such that B is a diagonal matrix and $A = SBS^{-1}$. Also find S^{-1} .

$$S = \begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

$$S^{-1} = \frac{1}{\det S} \begin{pmatrix} -1 & -1 \\ 2 & 1 \end{pmatrix}, \quad \det S = 1(-1) - 1(-2) = 1.$$

$$S^{-1} = \begin{pmatrix} -1 & -1 \\ 2 & 1 \end{pmatrix}, \quad A = SBS^{-1}.$$

$$\boxed{S = \begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad S^{-1} = \begin{pmatrix} -1 & -1 \\ 2 & 1 \end{pmatrix}.$$

f) Use part (e) to find A^n .

$A^n = SB^nS^{-1}$ with $B^n = \text{diag}(2^n, 3^n)$.

$$A^n = \begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} 2^n & 0 \\ 0 & 3^n \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 2 & 1 \end{pmatrix}.$$

$$A^n = \begin{pmatrix} -2^n + 2 \cdot 3^n & -2^n + 3^n \\ 2^{n+1} - 2 \cdot 3^n & 2^{n+1} - 3^n \end{pmatrix}.$$

$$\boxed{A^n = \begin{pmatrix} -2^n + 2 \cdot 3^n & -2^n + 3^n \\ 2^{n+1} - 2 \cdot 3^n & 2^{n+1} - 3^n \end{pmatrix}.$$

2. Let

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Answer questions (a)–(e) (from Problem 1) for this A . You do not have to do part (f) and you do not have to find S^{-1} .

$$\chi_A(\lambda) = (\lambda - 2)(\lambda + 1)^2, \quad \lambda = 2, -1, -1.$$

$$E_2 = \text{span}\{(1, 1, 1)^\top\}, \quad E_{-1} = \text{span}\{(-1, 1, 0)^\top, (-1, 0, 1)^\top\}.$$

$$S = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

$$\boxed{A = SBS^{-1}, \quad A \text{ is diagonalizable.}}$$

3. Let $a \neq 0$.

(a) Explain why the matrix

$$A = \begin{pmatrix} 0 & a \\ a & 0 \end{pmatrix}$$

is always diagonalizable. You can do this by just finding the eigenvalues and quoting a theorem.

$$\det \begin{pmatrix} -\lambda & a \\ a & -\lambda \end{pmatrix} = \lambda^2 - a^2 = (\lambda - a)(\lambda + a).$$

eigenvalues are $\lambda = a, -a$. Distinct eigenvalues $\Rightarrow$ diagonalizable.

$$\boxed{\text{Distinct eigenvalues } a, -a \Rightarrow A \text{ is diagonalizable.}}$$

(b) Explain why the matrix

$$A = \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix}$$

is never diagonalizable.

$$\det \begin{pmatrix} -\lambda & a \\ -a & -\lambda \end{pmatrix} = \lambda^2 + a^2.$$

This gives $\lambda = \pm ai$, which are complex, not real. Hence A has no real eigenvalues and is not diagonalizable over $\mathbb{R}$.

$$\boxed{A \text{ has no real eigenvalues} \Rightarrow \text{not diagonalizable over } \mathbb{R}.}$$